NATIONAL OSTEOPOROSIS FOUNDATION

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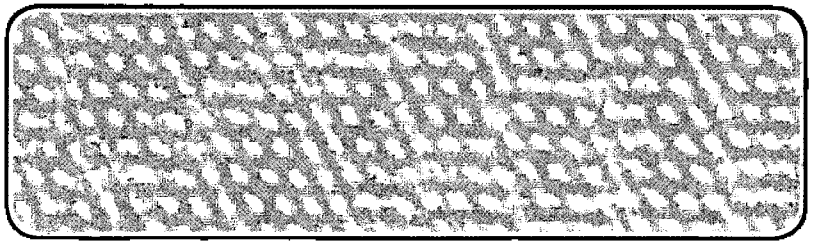

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NATIONAL OSTEOPOROSIS FOUNDATION

A Guide to Osteoporosis medicines

A Guide to Osteoporosis Medicines can help you understand osteoporosis and your treatment choices. It gives you information about the osteoporosis medicines approved by the U.S. Food and Drug Administration (FDA). This brochure also has information to help you make decisions about your treatment.

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Established in 1984, the National Osteoporosis Foundation (NOF) is the nation's leading voluntary health organization solely dedicated to osteoporosis and bone health.

National Osteoporosis Foundation's Vision
To make bone health a reality and a lifelong priority for all individuals.

National Osteoporosis Foundation's Mission
To prevent osteoporosis and related fractures, to promote lifelong bone health, to help improve the lives of those affected by osteoporosis and to find a cure through programs of awareness, advocacy, public and health professional education and research.

The National Osteoporosis Foundation (NOF) offers the following publications and services:

- **Free** brochures and handouts for people of all ages and backgrounds, including information in Spanish
- **Free** osteoporosis support groups
- **Free** online osteoporosis community
- Materials for community events and health fairs
- 100-page handbook, *Boning Up on Osteoporosis,* with up-to-date information
- Exercise video/DVD
- Online walking program and other special events
- Individual memberships that include a quarterly newsletter and much more!

To request information, become an NOF member or learn more about these programs and services, contact NOF at:

(800) 231-4222 (Toll-free)
(202) 223-2226 (Main)
(202) 223-2237 (Fax)
E-mail: request@nof.org
Web site: www.nof.org

For More Information

Medicines. For additional information on medicines, the following Web sites may be helpful:

- DrugDigest—www.drugdigest.org
- Food and Drug Administration (FDA)—www.fda.gov
- MedlinePlus—www.nlm.nih.gov/medlineplus/druginformation.html
- Physicians' Desk Reference (PDR)—www.pdrhealth.com

The following Web sites have information on the results of clinical studies, including those on medicines:

- www.clinicalstudyresults.org
- www.pubmed.gov

My Medicine Record. This form helps you keep track of your prescription medicines, over-the-counter medicines and dietary supplements. You can use "My Medicine Record" to share information with your healthcare providers. It was developed by the Food and Drug Administration (FDA). The FDA suggests taking it with you when you visit your healthcare provider, pharmacy or hospital.

- The record can be printed from the FDA Web site at www.fda.gov/cder/consumerinfo/my_medicine_record.htm.
- Other information, including safety alerts, is available on the FDA Web site at www.fda.gov.
- The toll-free FDA phone number is (888) 463-6332.

MUST for Seniors™. This program helps older adults and caregivers learn how to safely use medicines, prevent mistakes and recognize common side effects. MUST stands for "Medication Use Safety Training." The National Council on Patient Information and Education (NCPIE) sponsors the program.

- The Web site is www.mustforseniors.org.
- Other information, including how to safely store and dispose of your medicines, is available on NCPIE's Web site at www.talkaboutrx.org.
- The NCPIE phone number is (301) 656-8565.

Note: The Web sites and phone numbers listed are current at the time of publication.

What Is Osteoporosis?

Osteoporosis is a disease that causes bones to become weak and break more easily. People don't feel their bones becoming weak. So, it's a disease that can sneak up on you. For some people, breaking a bone is the first clue that they have osteoporosis. Breaking a bone is serious, especially when you're older. Today, we know how to prevent, diagnose and treat osteoporosis. We also know many of the things that increase your chance of having the disease and the broken bones it can cause. These are called risk factors for osteoporosis. Major ones include:

- Not getting enough calcium and vitamin D
- Having a parent who had osteoporosis or broke a bone as an adult
- Being small and thin
- Taking certain medicines (examples are steroid medicines, certain treatments for cancer and some antiseizure medicines)
- Smoking

Although there is no cure for osteoporosis, it can be treated. You can slow or stop it from getting worse. In some cases, your bones may even improve.

How Bones Grow and Change

Some people think of bones as hard and lifeless. But, your bones are actually living, growing tissue. Did you know that throughout your life you constantly lose old bone while you make new bone at the same time? This is the **bone remodeling cycle,** and there are two parts to it:

- **Bone resorption.** You have special cells in your bones that help break down and remove old bone. These cells are called osteoclasts.
- **Bone formation.** You also have special cells in your bones that help form new bone. This new bone replaces the old bone that is removed. These cells are called osteoblasts.

Osteoporosis happens when you lose too much old bone, form too little new bone or both. Although you can improve your bone health at any age, the best time to build strong, healthy bones is when you're young. The more bone you build when you're young, the less likely you are to break a bone or get osteoporosis later in life. But, it's never too late at any age to take steps to protect your bones.

If your healthcare provider hasn't talked to you about your bone health, it's time for you to bring it up. Depending on your age and risk factors for osteoporosis, your healthcare provider may recommend a bone mineral density (BMD) test. It lets you know the amount of bone you have in your hip, spine or other bones. A BMD test is how to find out if you have osteoporosis.

Special Note: The information about osteoporosis medicines in this brochure is current at the time of publication. It is intended as general information only. You should not depend on it for any specific diagnosis or treatment. This information is not a recommendation by the National Osteoporosis Foundation (NOF) of any particular medicine or company that makes an osteoporosis medicine.

For more detailed information about osteoporosis medicines, you should speak with your doctor, pharmacist or other healthcare provider. You can also read the package inserts, which are available online and at pharmacies.

Companies That Make Osteoporosis Medicines

Generic (Brand Name)	Company and Phone Number	FDA Approval Date
Alendronate (Fosamax®, Fosamax Plus D™)	Merck (800) 672-6372	1995
Calcitonin (Fortical®, Miacalcin®)	▪ Fortical® Nasal Spray–Upsher-Smith (800) 654-2299 ▪ Miacalcin® Nasal Spray or Injection–Novartis (888) 669-6682	2005 1995
Estrogen Therapy (ET) and Hormone Therapy (HT) Multiple brand names are available. For example: ET–Estrace® and Premarin® HT–FemHrt® and Prempro®	Too many to provide complete list. Talk to your pharmacist to get information on a specific brand.	
Ibandronate (Boniva®)	Roche (800) 526-6367	2005– Monthly Oral Dose 2006– Quarterly IV Dose
Raloxifene (Evista®)	Eli Lilly (800) 545-5979	1997
Risedronate (Actonel®, Actonel® with Calcium)	Procter & Gamble (800) 448-4878	2000
Teriparatide (Forteo®)	Eli Lilly (800) 545-5979	2002
Zoledronic Acid (Reclast®)	Novartis (888) 669-6682	2007

Exercise. Your bones get stronger when you make them work. People who are bedridden or do not exercise are at high risk for osteoporosis. You should exercise about 30 minutes on most days of the week. The two types of exercises that help keep your bones strong and healthy include:

- **Weight-bearing exercises.** Examples are walking, dancing, hiking, jogging, elliptical training machines and aerobics.
- **Muscle-strengthening exercises.** Examples are lifting weights, using elastic exercise bands and using weight machines.

Check with your healthcare provider before beginning a new exercise program. If you have osteoporosis or an increased chance of breaking a bone, your healthcare provider may also recommend that you see a physical therapist. A physical therapist can teach you safe ways to move and exercise.

Reviewing Your Treatment Plan

If you take an osteoporosis medicine, you should review your treatment plan every year with your healthcare provider. If you have been taking an osteoporosis medicine for five years, discuss the benefits of continuing it. People who are not at high risk of breaking a bone may be able to take a "drug holiday" after five years of treatment. This means that you stop taking your osteoporosis medicine but continue to see your healthcare provider to monitor your bone health.

If you are at high risk of breaking a bone, then you may benefit by staying on your osteoporosis medicine. Other people may benefit from switching to a different medicine. Your healthcare provider is the best person to guide you about whether you should start, continue, switch or stop an osteoporosis medicine. Again, it is always important to look at both the benefits and risks of any medicine when making a decision about your treatment.

What More You Can Do

In order for your medicine to work, you still need to get enough calcium, vitamin D and exercise.

Calcium. Getting enough calcium helps build strong bones when you're younger and helps keep them strong later in life. If you get the recommended amount of calcium from the foods you eat, you don't need to take a calcium supplement. If you are not sure, talk to your healthcare provider to find out if you need to take a calcium supplement and how much to take.

Adults under age 50
need 1,000 mg of calcium and 400–800 IU of vitamin D every day.
Adults age 50 and older
need 1,200 mg of calcium and 800–1,000 IU of vitamin D every day.

Vitamin D. Your body needs vitamin D to absorb calcium. You can get small amounts of vitamin D from a few foods. You can also get vitamin D from the sun, but you need to protect your skin from too much sun. Many people need to take a supplement. Ask your healthcare provider if you're getting enough vitamin D and whether you should have a simple blood test to check your vitamin D level.

When to Take an Osteoporosis Medicine

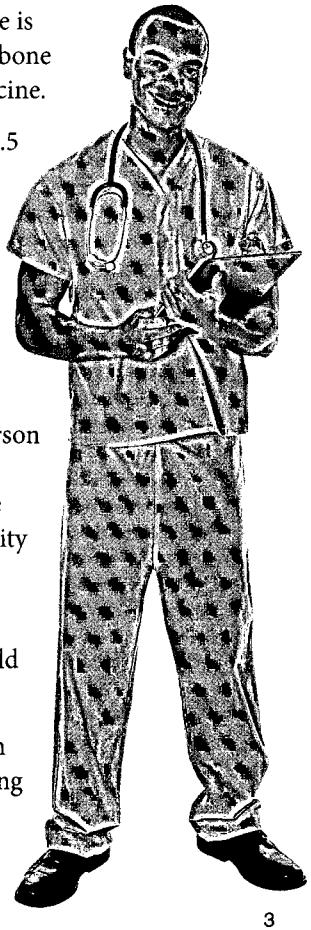
Your healthcare provider will look at certain information to decide what you can do to prevent or treat osteoporosis. This includes your risk factors for osteoporosis, your likelihood of breaking a bone, your medical history, your current health and the results of your BMD test.

The results of your BMD test will show a T-score for each bone that is tested. Healthcare providers use the lowest T-score to diagnose osteoporosis in women going through menopause, postmenopausal women and men age 50 and older. It shows how much your bone density is above or below normal compared to an average 30-year-old healthy adult.

Below are treatment guidelines for postmenopausal women and for men age 50 and older based on BMD test results:

- **Normal bone density.** A T-score of -1 and above is normal bone density. Most people with normal bone density don't need to take an osteoporosis medicine.
- **Low bone density.** A T-score between -1 and -2.5 is low bone density. It's also called **osteopenia.** People with low bone density are more likely to get osteoporosis. They should consider taking an osteoporosis medicine when they have certain risk factors that suggest an increased chance of breaking a bone.

  Healthcare providers can use a Web-based tool called FRAX® to figure out the chance that a person will break a hip or other major bone within the next 10 years. This information helps healthcare providers decide if a person with low bone density should take an osteoporosis medicine.
- **Osteoporosis.** A T-score of -2.5 and below is osteoporosis. All people with osteoporosis should consider taking an osteoporosis medicine.
- **Certain broken bones.** People who have broken a hip or bones in the spine should consider taking an osteoporosis medicine.



What to Consider When Choosing a Medicine

There is a lot to think about when choosing an osteoporosis medicine. You and your healthcare provider may want to look at:

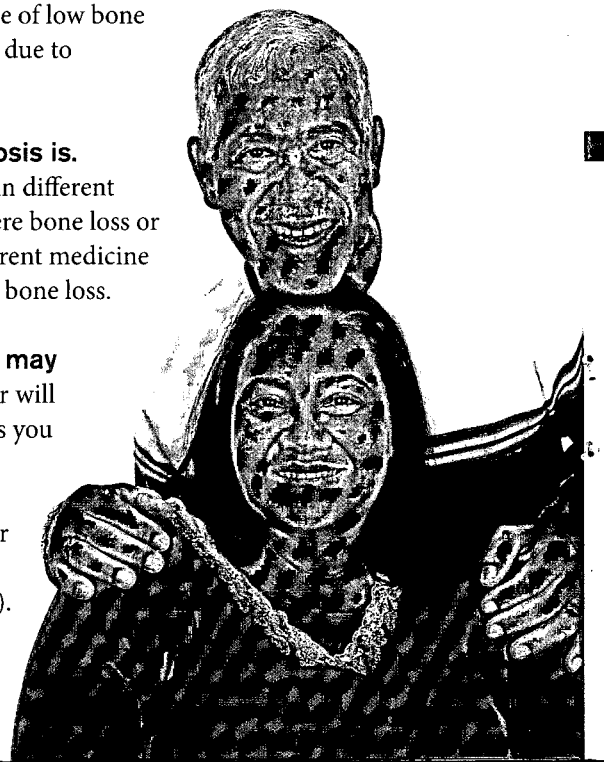
Your gender. While all osteoporosis medicines are approved for women, only four are approved for men. The ones approved for men are alendronate (Fosamax®), risedronate (Actonel®), teriparatide (Forteo®) and zoledronic acid (Reclast®).

Your age. Some medicines may be more appropriate for women who have recently gone through menopause while others are more appropriate for older women. You should talk to your healthcare provider about what medicine is best for you at this time.

If you have not reached menopause. Most of the osteoporosis medicines are not approved for premenopausal women. But, three osteoporosis medicines are approved for the prevention or treatment of osteoporosis in premenopausal women as a result of taking steroid medicines for a long time. In very rare cases, a healthcare provider may recommend that a premenopausal woman consider taking an osteoporosis medicine for other reasons. Examples include when a woman breaks a bone because of low bone density or has severe bone loss due to a medical condition.

How severe your osteoporosis is. Osteoporosis medicines work in different ways. A person with more severe bone loss or a broken bone may take a different medicine than a person with only minor bone loss.

Other health problems you may have. Your healthcare provider will consider other health problems you have when recommending a medicine. If you have had breast cancer or blood clots, for example, you should not take estrogen or raloxifene (Evista®).



How to Know If Your Medicine Is Working

To find out how well your treatment is working, your healthcare provider might repeat your bone mineral density (BMD) test every two years. Healthcare providers may have some patients repeat a BMD test after one year. In some cases, healthcare providers will also use lab tests to see if patients are losing bone faster than normal. While there is no easy way to measure improvement in bone structure, current research is being done on this issue.

For some people, an osteoporosis medicine doesn't work as well when they have a disease or condition that causes bone loss. Some examples are vitamin D deficiency, celiac disease and rheumatoid arthritis. Your healthcare provider may check for diseases and conditions that can cause bone loss before you start taking an osteoporosis medicine. Or, your healthcare provider may do this if your medicine is not working as it should. Sometimes treating a health problem that causes bone loss can improve your bone health.

How Long to Take an Osteoporosis Medicine

At this time, healthcare providers don't know for sure how long most osteoporosis medicines stay safe and effective to take. They also don't know how long these medicines continue to be helpful after you stop taking them. Research suggests that the benefits of bisphosphonate medicines may continue for several years or longer after you stop taking them. This is because these medicines stay in the bones for a long time.

There have been no studies on any of the osteoporosis medicines that have lasted more than 10 years. Longer research studies are needed on all of the osteoporosis medicines to improve our understanding of the safety and benefits of using these medicines over a period of many years.

What to Expect From Your Medicine

With antiresorptive medicines, the goal of treatment is to prevent more bone loss and reduce the risk of breaking bones in the future. Response to treatment with these medicines is considered good if:

- Your bone density stays the same or improves
- You don't break any bones

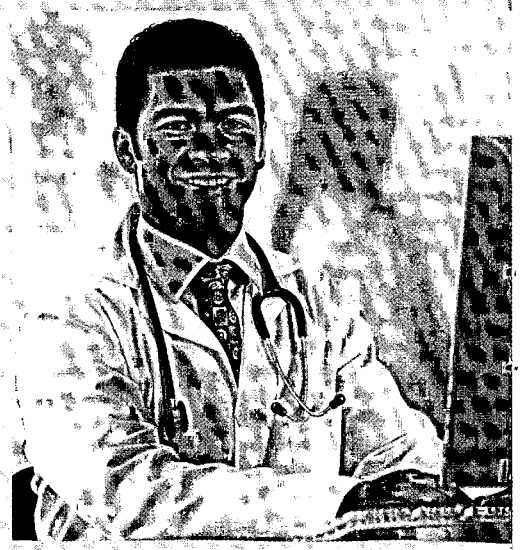
With the one anabolic medicine, the goal of treatment is to build new bone, increase bone density, repair tiny defects in bone and reduce the risk of breaking bones. Response to treatment is considered good if:

- Your rate of making new bone increases
- Your bone density improves
- You don't break any bones

Staying With Your Treatment Plan

When you take an osteoporosis medicine, you will not feel your bones getting stronger. This can make it hard to stay on a treatment plan. But, it's important to take your medicine if you want it to work. You should take it just as your healthcare provider prescribed it, and you must remember to take it. When you have questions about your medicines, be sure to speak with your healthcare provider or pharmacist.

If you decide that a treatment is not right for you, don't just stop taking the medicine. First, talk with your healthcare provider about your concerns. When prescriptions are not filled, or if they are forgotten, taken incorrectly or stopped early, your health condition may not improve and could get worse. Your healthcare provider may find it difficult to figure out why you aren't getting better. This can lead to extra tests, prescriptions, costs and broken bones that could have been prevented by taking the medicine as directed.



If you've had radiation treatment involving your bones, you should not take teriparatide (Forteo®).

> Keep in mind that how well a medicine works, or what side effects it may have, will not be the same for everyone.

Personal preference. Do you prefer a pill, liquid or intravenous (IV) medicine or one that is given as a nasal spray or an injection? Does it work better for you to take your medicine every day, once a week, once a month, several times a year or even once a year? Any of these could influence your treatment decision.

Medicines to Prevent or Treat Osteoporosis

The U.S. Food and Drug Administration (FDA) has approved a number of medicines to prevent or treat osteoporosis. These medicines fall into two main types:

Antiresorptives. In the bone remodeling cycle, antiresorptive medicines slow the breakdown of bone. This helps prevent bone loss. When you first start taking these medicines, you stop breaking down bone as quickly as you did before, but you still make new bone at a normal rate. So, your bone density might even increase. These medicines also lower your risk of breaking bones. The antiresorptives are:

- Bisphosphonates
- Calcitonins
- Estrogen agonists/antagonists
- Estrogen therapy (ET) and hormone therapy (HT)

Anabolics. In the bone remodeling cycle, anabolic medicines help your body make new bone more quickly. They increase your bone density and lower your risk of breaking bones. Only one type of anabolic medicine has been approved for osteoporosis. It is:

- Parathyroid hormone

The charts on pages 7 and 10 provide some facts about osteoporosis medicines. The next sections include more detailed information about these medicines.

Antiresorptive Medicines

Antiresorptive medicines slow the breakdown of bone. The following information is about the four types of antiresorptive medicines.

Bisphosphonates

There are four bisphosphonate medicines available. They are:

- Alendronate (Fosamax®), also available as a generic medicine
- Ibandronate (Boniva®)
- Risedronate (Actonel®)
- Zoledronic acid (Reclast®)

The bisphosphonates can increase bone density. They lower the risk of breaking bones in the spine. Alendronate, risedronate and zoledronic acid lower the risk of breaking a hip and other bones. For people who have recently broken a hip, zoledronic acid lowers the risk of new broken bones.

All four bisphosphonate medicines are approved for postmenopausal women. Alendronate, risedronate and zoledronic acid are approved for men. Ibandronate is for women only. Some bisphosphonates are for both the prevention and treatment of osteoporosis and others are for treatment only. See the chart on page 10 for more information.

For women and men, risedronate and zoledronic acid are also approved to prevent and treat osteoporosis caused by taking steroid medicines, such as cortisone and prednisone. Alendronate is approved to treat osteoporosis caused by these medicines.

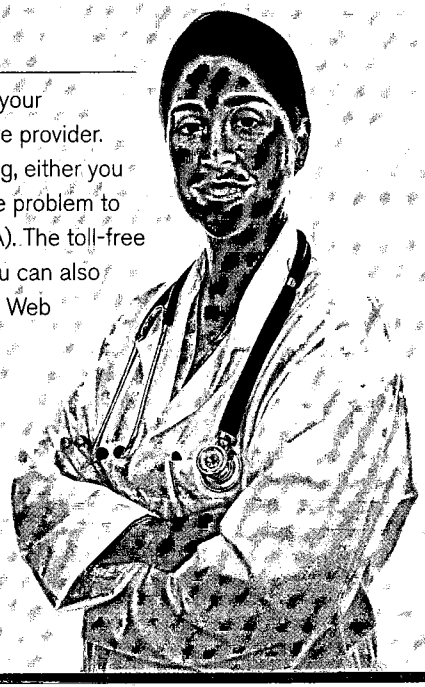
How to take bisphosphonates. Bisphosphonates are available in the following forms:

- **Tablets.** Alendronate, ibandronate and risedronate are available as tablets taken either once a day, once a week, twice a month or once a month. You take one tablet first thing in the morning after waking up and on an empty stomach. The tablet must be swallowed whole with 6 to 8 ounces of plain water (no other liquid), at least 30 minutes before having anything to eat or drink. When you take an ibandronate tablet, you need

Reporting Side Effects

If you think you're having a side effect from your medicine, it's important to tell your healthcare provider. For serious reactions or problems with a drug, either you or your healthcare provider should report the problem to the U.S. Food and Drug Administration (FDA). The toll-free number for the FDA is (800) 332-1088. You can also complete a report online by visiting the FDA Web site at www.fda.gov/medwatch.

You may also want to report serious side effects and problems to the company that makes the medicine. The chart on page 19 provides the phone numbers of these companies and the date each medicine was approved by the FDA as an osteoporosis medicine.



that has spread to the bone. Also, people who have certain abnormal blood tests, including increased calcium levels, should not take this medicine.

Special concerns about parathyroid hormone. In rat studies, the drug was given in very high doses over a long period of time and caused a rare bone cancer. This has not been seen in humans. There has been no evidence of an increased risk of this cancer in humans taking teriparatide.

Taking Two Medicines at the Same Time

Sometimes healthcare providers recommend taking two different osteoporosis medicines at the same time. This usually means taking a bisphosphonate and a medicine that is not a bisphosphonate. Taking two osteoporosis medicines at the same time may provide a small increase in bone density compared to only taking one. It is not known if this provides greater protection from breaking bones. Also, it's possible that taking two medicines may cause more side effects. If your healthcare provider suggests adding a second medicine, you will need to decide if the possible gains are worth the added cost and possible side effects.

Anabolic Medicines

Anabolic medicines help build new bone. The following information is about the one type of anabolic medicine.

Parathyroid Hormone

Teriparatide (Forteo®) is a type of parathyroid hormone.
It is the only anabolic or bone-building medicine currently approved for osteoporosis. The FDA has approved it to treat osteoporosis in postmenopausal women and in men.

Teriparatide lowers the risk of breaking bones in the spine and breaking other bones. This medicine builds new bone and greatly increases bone density, especially in the spine.

You may consider taking teriparatide if you have broken a bone due to osteoporosis and have very low bone density, such as a T-score lower than -3.0. Teriparatide may also be a choice if you continue to lose bone density or break a bone while being treated with another osteoporosis medicine.

How to take parathyroid hormone. Teriparatide is given by a daily self-injection. The FDA has approved its use for up to two years only. At the end of two years, to retain the benefits of treatment with teriparatide, most experts recommend that patients start an antiresorptive medicine. This is usually a bisphosphonate.

Side effects of parathyroid hormone. Side effects can include leg cramps and dizziness. Small increases of calcium in the blood and urine can occur, but this medicine does not increase the risk of kidney stones.

People with certain conditions should not take this medicine. This includes people with Paget's disease of bone and people who have had radiation treatment that involved the bones. It also should not be given to people with metabolic bone diseases such as hyperparathyroidism and those with cancer

Medicines to Prevent or Treat Osteoporosis

Class	Drug	Brand Name	Form	Frequency
Antiresorptives	**Bisphosphonates**			
	Alendronate	Generic Alendronate and Fosamax®	Oral (tablet)	Daily/Weekly
	Alendronate	Fosamax Plus D™ (with 2,800 IU or 5,600 IU of Vitamin D_3)	Oral (tablet)	Weekly
	Alendronate	Fosamax®	Oral (liquid solution)	Weekly
	Ibandronate	Boniva®	Oral (tablet)	Monthly
	Ibandronate	Boniva®	Intravenous (IV) injection	Four Times per Year
	Risedronate	Actonel®	Oral (tablet)	Daily/Weekly/ Twice Monthly/ Monthly
	Risedronate	Actonel® with Calcium	Oral (tablet)	Weekly
	Zoledronic Acid	Reclast®	Intravenous (IV) infusion	One Time per Year
	Calcitonins			
	Calcitonin	Fortical®	Nasal spray	Daily
	Calcitonin	Miacalcin®	Nasal spray	Daily
	Calcitonin	Miacalcin®	Injection	Varies
	Estrogen Agonists/Antagonists			
	Raloxifene	Evista®	Oral (tablet)	Daily
	Estrogen Therapy (ET) and Hormone Therapy (HT)*			
	Estrogen	Multiple Brands	Oral (tablet)	Daily
	Estrogen	Multiple Brands	Transdermal (skin patch)	Twice Weekly/ Weekly
Anabolics	**Parathyroid Hormone**			
	Teriparatide	Forteo®	Subcutaneous injection (under the skin)	Daily

*Estrogen is also available in other preparations including a vaginal ring, cream, injection and oral tablet taken sublingually (under the tongue). The vaginal preparations do not provide much bone protection.

Benefits and Risks

All medicines have benefits and risks of side effects. You should talk to your doctor or other healthcare provider about both the benefits and risks of a medicine before taking it. If you have osteoporosis or an increased chance of breaking a bone, ask your healthcare provider about the treatment choices available to you.

to wait at least 60 minutes before eating or drinking. You must remain upright (sitting, standing or walking) during this 30-minute or 60-minute period.

For Actonel® with calcium, the weekly Actonel® tablet is available with separate calcium carbonate tablets. The calcium tablet is **not** taken at the same time as the Actonel® tablet because calcium prevents the absorption and activity of Actonel®.

For Fosamax Plus D™, the weekly Fosamax® tablet contains both alendronate and vitamin D. The vitamin D is not in a separate tablet or capsule.

- **Liquid.** Alendronate is also available as a liquid (oral) solution. This oral solution is taken once a week, first thing in the morning on an empty stomach. You need to drink the entire bottle of solution. Right after you drink the solution, you must drink at least 2 ounces of plain water. After drinking the solution and 2 ounces of plain water, you must remain upright (sitting, standing or walking) for at least 30 minutes with no food or drink during this time.
- **Intravenous (IV).** Ibandronate is available as an intravenous (IV) injection given every three months. A healthcare professional gives the IV injection in a doctor's office or other outpatient setting. It takes less than a minute to inject the medicine. You should have a blood test to check your kidneys before each IV injection. Some healthcare providers may also check blood calcium and vitamin D levels.

  Zoledronic acid is the first and only once-a-year osteoporosis medicine. A healthcare professional gives zoledronic acid as an intravenous (IV) infusion in a doctor's office or other outpatient setting. It takes at least

bones in the spine and breaking a hip and other bones. ET and HT also help relieve menopausal symptoms. Many brands and generic forms are available.

How to take ET and HT. ET and HT are available as a tablet or skin (transdermal) patch and in other forms. Estrogen and hormone medicines come in many different doses.

Side effects of ET and HT. Taking estrogen by itself increases a woman's risk for cancer of the uterus. To reduce this risk, women with a uterus are given estrogen with progesterone (HT). Estrogen alone (ET) is given to women who have had their uterus removed. Side effects may include vaginal bleeding, breast tenderness and gallbladder disease.

Although estrogen can reduce bone loss, large studies have found that it also slightly increases a woman's risk of stroke, blood clots, breast cancer and other problems. As a result, the FDA recommends that if you need a medicine for osteoporosis you look at other medicines first before taking ET or HT. Because ET and HT can have serious risks, you should talk to your healthcare provider to see if the benefits outweigh the risks. If you decide to take ET or HT, you should take the lowest possible dose for the shortest period of time.

Special Concerns About ET and HT

The Woman's Health Initiative (WHI) study found that one type of HT called Prempro® reduced the risk of breaking a hip and other bones and reduced the risk of getting colon cancer. In the study, it was given to women who on average were more than 10 years past menopause. It was associated with a slight increase in the risk of breast cancer, strokes, heart attacks, blood clots and cognitive (mental) decline. Although ET was associated with a similar increase in the risk of strokes, blood clots and mental decline, it did not increase the risk of breast cancer or heart attacks. Recent studies suggest that the risks of HT may be lower in women closer to menopause.

Studies show that raloxifene decreases the risk of the most common type of breast cancer that occurs in postmenopausal women. The FDA has approved it to decrease the risk of breast cancer in women with osteoporosis and even in women without osteoporosis who are at high risk of breast cancer.

How to take estrogen agonists/antagonists. For both the prevention and treatment of osteoporosis, raloxifene is taken every day as a tablet with or without meals.

Side effects of estrogen agonists/antagonists. Side effects with raloxifene can include blood clots, hot flashes and leg cramps. Other side effects include swelling and temporary flu-like symptoms. Raloxifene is not associated with problems of the uterus or ovaries and does not affect cognitive (mental) function. Because this medicine can cause hot flashes, it may not be the best choice for women who have recently gone through menopause and are still having menopausal symptoms.

Women who have had blood clots or breast cancer should not take this medicine. They should also not take it if they have an increased risk for stroke. This includes women who:

- Have already had a stroke
- Have transient ischemic attacks (TIAs), which are sometimes called "mini strokes" and can come and go quickly
- Have atrial fibrillation, which is a type of serious irregular heart beat
- Have uncontrolled high blood pressure
- Are smokers

Estrogen Therapy (ET) and Hormone Therapy (HT)

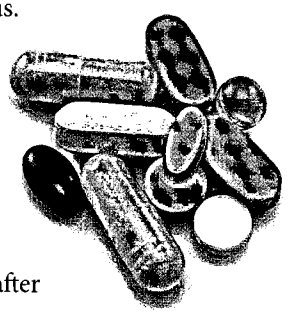
Estrogen therapy (ET) and hormone therapy (HT) are for women only. The terms refer to therapy with estrogen alone (ET) and estrogen given with progesterone (HT). ET and HT are approved to prevent osteoporosis in postmenopausal women. Estrogen is a hormone that protects bone. It can increase bone density. It lowers the risk of breaking

15 minutes for the yearly infusion. You should eat as you normally would, but you need to drink two glasses of water or another liquid within a few hours before receiving zoledronic acid. You will need to have two blood tests before each IV dose. One is a test to check your kidneys. The other is a test to check calcium and possibly other mineral levels in the blood.

Side effects of bisphosphonates. Bisphosphonates are not recommended for people with severe kidney disease or low blood calcium. People with certain problems of the esophagus may not be able to take the oral tablets. The esophagus is the tube connecting the throat to the stomach.

The side effects of **all bisphosphonates** can include bone, joint and muscle pain. Eye inflammation and osteonecrosis of the jaw (ONJ) are rare side effects of all bisphosphonate medicines.

The more common side effects of **bisphosphonates taken as a tablet or liquid** include problems with the digestive system, such as trouble swallowing, heartburn and irritation of the esophagus.



Side effects that can occur shortly after receiving **IV bisphosphonates** include flu-like symptoms, fever and headaches. These generally stop within two to three days and usually only happen after the first IV infusion. To help prevent or relieve these symptoms, patients may be given acetaminophen or another over-the-counter medicine either before or after the IV infusion.

Special concerns about bisphosphonates. The National Osteoporosis Foundation (NOF) recognizes that certain serious conditions associated with bisphosphonate medicines may cause concern for people taking them. There have been reports of atrial fibrillation, osteonecrosis of the jaw (ONJ), severe pain and unusual broken bones in the thigh. While these conditions are serious, the number of people with them remains extremely small compared to the number of people who have taken these medicines. It is not yet known whether the medicines are truly causing these problems, in part because the problems are so rare.

Based on information that is currently available, NOF believes that for most people taking bisphosphonate medicines, the benefits outweigh the risks of developing one of these unusual but serious conditions. These medicines

Osteoporosis Medicines

FDA Approved For:	Postmenopausal Women	Men	Prevents Osteoporosis	Treats Osteoporosis	Prevents Osteoporosis Caused by Steroid Medicines	Treats Osteoporosis Caused by Steroid Medicines
Bisphosphonates						
Alendronate (Fosamax®, Fosamax Plus D™)	■	■*	■	■		■
Ibandronate (Boniva®)	■		■	■†		
Risedronate (Actonel®, Actonel® with Calcium)	■	■*	■	■	■	■
Zoledronic Acid (Reclast®)	■	■		■	■	■
Calcitonins						
Calcitonin (Fortical®)	■			■		
Calcitonin (Miacalcin®)	■			■		
Estrogen Agonists/Antagonists						
Raloxifene (Evista®)	■		■	■		
Estrogen Therapy (ET) and Hormone Therapy (HT)						
Many brands	■		■			
Parathyroid Hormone						
Teriparatide (Forteo®)	■	■		■		

*For men, alendronate and risedronate are approved for treatment only.
† Ibandronate as an intravenous (IV) injection is approved for treatment only.

play an important role in stopping bone loss and preventing broken bones. You should discuss your concerns about these possible side effects with your healthcare provider.

Calcitonins

Calcitonin is a hormone that affects calcium and bone. Calcitonin medicines are approved to treat osteoporosis in women who are at least five years past menopause. These medicines are available as either Fortical® or Miacalcin®.

Calcitonins can increase bone density in the spine. They lower the risk of breaking bones in the spine, but they do not lower the risk of breaking a hip and other bones.

How to take calcitonins. Both Fortical® and Miacalcin® are available as a spray used in the nose. The nasal spray is usually used once a day, with the patient switching nostrils every day. Miacalcin® is also available as an injection. It is given under the skin or into a muscle, usually once a day or once every other day.

Side effects of calcitonins. Common side effects with nasal calcitonin are runny nose, headache, back pain and nosebleeds. Injectable calcitonin may cause an allergic reaction and unpleasant side effects, such as flushing of the face and hands, needing to urinate often, nausea and a skin rash.

Estrogen Agonists/Antagonists

Estrogen agonists/antagonists are for postmenopausal women only. They used to be called selective estrogen receptor modulators (SERMs). They were developed to provide the benefits of estrogen therapy without the risks. They are not a hormone or estrogen.

The only estrogen agonist/antagonist approved to prevent or treat osteoporosis is raloxifene (Evista®). It can increase bone density. While it lowers the risk of breaking bones in the spine, the studies so far do not show that it lowers the risk of breaking a hip and other bones.